From Long-Form Dialogue to TTS Corpora

Problem, Four-Stage Pipeline, and Related Work

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October 2026

Abstract

Modern text-to-speech systems learn from corpora of short, clean, transcribed utterances, but the raw material is usually long-form dialogue that was never recorded for that purpose. This document introduces the problem and the four-stage pipeline used by the present work: collection, slicing, semantic enrichment, and model training with synthesis evaluation. We describe the failure modes that constrain corpus construction—phoneme clipping, transient clicks, mid-sentence over-segmentation, and run-on utterances—and the design principles that recur across stages: a shared data contract, original-rate coordinates, non-overlap, a duration contract, terminal conditioning, and reproducibility. We then review related work in voice activity detection, ASR word timing, TTS corpora and systems, and segmentation evaluation, and explain how the phase-specific documents fit together.

1 The corpus-construction problem

A text-to-speech (TTS) model is only as good as the corpus it is trained on. Modern systems range from single-speaker architectures trained on many hours of studio speech to few-shot and zero-shot systems that adapt from a few minutes of reference audio, but in every case training requires utterances that are short enough to fit an attention window, long enough to carry prosody, acoustically clean at both edges, and paired with an accurate transcript. The raw material rarely has these properties. Game dialogue, animation, audiobooks, and archival recordings arrive as continuous waveforms with music beds, sound effects, long pauses, overlapping speakers, and no sentence-level annotation.

A first implementation often applies a fixed-duration window or a conventional voice activity detector (VAD). Both approaches fail in predictable ways. A fixed window cuts through phonemes and ignores syntax; a frame-level VAD answers "is this frame speech?" but not "is this a trainable utterance?". The gap between the two questions is the engineering problem addressed by this project.

Four failure modes dominate corpus preparation and recur as design requirements throughout the pipeline:

1. **Unvoiced-phoneme clipping.** Fricatives and plosives have low root-mean-square (RMS) energy but high-frequency content. A pure energy gate removes them, so an utterance begins or ends mid-phoneme.
2. **Transient clicks.** A cut at a sample with non-zero amplitude creates a step discontinuity whose spectrum decays as $O(1/f)$. After neural vocoding it is audible as a click. The first and last samples of a training clip must be driven smoothly to zero.
3. **Mid-sentence over-segmentation.** A dramatic pause of 300–500 ms is shorter than a listener's notion of a sentence boundary but longer than many VAD hangover windows. Cutting there fragments a clause into two unusable pieces.

4. **Run-on utterances.** Monologues longer than roughly 12 s increase memory pressure and the risk of attention collapse or hallucination during training. The corpus needs a hard ceiling and a balanced way to subdivide long regions.

The project therefore treats corpus construction as a sequence of constrained transformations rather than as a single model. The first transformation proposes a set of half-open intervals that jointly satisfy ordering, non-overlap, a duration contract, and edge-cleanliness conditions; later stages attach meaning to those intervals without changing their coordinates. Because the constraints conflict, the architecture exposes several interchangeable strategies rather than declaring a single "best" segmenter.

2 Related work

2.1 Voice activity detection and endpointing

Classical endpoint detection uses short-time energy and zero-crossing rate with explicit hangover rules [1]. Statistical VAD replaces the fixed threshold with a likelihood ratio between speech and noise models; the Gaussian-mixture approach of Sohn et al. remains a standard reference [2]. WebRTC's VAD applies a similar sub-band likelihood ratio in a small, real-time C implementation [3], while Silero VAD uses a recurrent neural network over log-mel features and is widely used as a robust pre-trained detector [4].

These systems answer a frame-level or region-level question: speech versus non-speech. They do not enforce a duration contract, resolve overlapping proposals, choose between a pause and a clause boundary, or guarantee that the cut sample is safe. Using a VAD directly for corpus preparation therefore transfers all of those decisions to an implicit post-processing step. The present work makes the post-processing explicit as a shared boundary algebra and evaluates the resulting intervals, not the frame labels.

2.2 ASR and word-level timing

Whisper is a Transformer speech model trained on large-scale weak supervision; it produces transcripts and word-level timestamps from attention [5]. WhisperX adds forced alignment and voice activity conditioning to improve timestamp accuracy on long-form audio [6], and Faster-Whisper provides an optimized CTranslate2 inference path [7]. Word timestamps make it possible to group words into clauses using punctuation and inter-word gaps, which is the basis of the semantic segmentation strategies used in this project.

The limitation is equally important: attention-derived timestamps are predictions, not measurements of acoustic silence. A word boundary can fall inside a fricative or a coarticulated transition. This motivates a hybrid strategy, which uses ASR only to decide *where* a clause ends and DSP to decide *which sample* inside the predicted pause is quietest.

2.3 TTS corpora and systems

LibriTTS provides multi-speaker English read speech with aligned transcripts and is a standard corpus for TTS research [8]. End-to-end systems such as VITS combine a conditional variational autoencoder, normalizing flows, and adversarial training [9]; few-shot systems such as GPT-SoVITS [10] and CosyVoice [11] can adapt to a new speaker from a small reference set. These systems differ in architecture and data requirements, but they share a preference for utterances in the range of a few seconds: too short and prosody is absent, too long and attention or memory becomes unstable. That preference is the origin of the soft duration window used throughout this project.

2.4 Segmentation quality and loudness

Reference-based segmentation evaluation matches predicted boundaries to ground truth under a tolerance and summarizes the result with precision, recall, and over-segmentation. The R-value of Räsänen et al. combines recall and over-segmentation into a single score and penalizes both missed and spurious boundaries [12]. Reference-free evaluation is harder: the present work uses edge energy, edge-energy

ratio, transient step jump, and a clean-boundary rate, and treats these as proxies rather than ground truth.

Loudness conditioning follows broadcast practice. EBU R128 specifies integrated loudness normalization and a maximum true-peak level [13], and ITU-R BS.1770-4 defines the K-weighting and gating algorithm used to measure it [14]. Conditioning is applied at materialization time, so normalization gain cannot feed back into boundary selection.

2.5 Positioning

The work described in this document differs from a standalone VAD or ASR segmenter in three ways. First, it treats the output as a constrained interval set with explicit invariants, rather than as a frame mask. Second, it keeps several heterogeneous strategies behind one interface and evaluates them on the same corpus with the same metrics, including a semantic-acoustic hybrid. Third, it separates corpus construction into phases with independent documents, so that preparation, enrichment, and model evaluation can be reproduced and extended without re-reading or rewriting the whole system.

3 A four-stage pipeline

Corpus construction is split into four decoupled stages (Table 1, Figure 1). Each stage has a clear input, output, and failure boundary, and each is specified by its own document so that it can be replaced or re-evaluated independently.

Table 1: The four corpus-construction stages and the scope of the phase documents.

Stage	Objective	Input → output	Primary risk
1	Collection	scripts, recordings → long-form raw audio and metadata	Licensing, source quality, missing transcripts
2	**Slicing**	long-form audio → intervals, conditioned clips, timing telemetry	Clipped phonemes, clicks, run-ons, fragmentation
3	Enrichment	clips → text, tags, dataset list	ASR hallucination, tag coverage, misalignment
4	Training and evaluation	corpus → trained model, MOS/SECS/CER report	Data leakage, overfitting, unstable perceptual scores

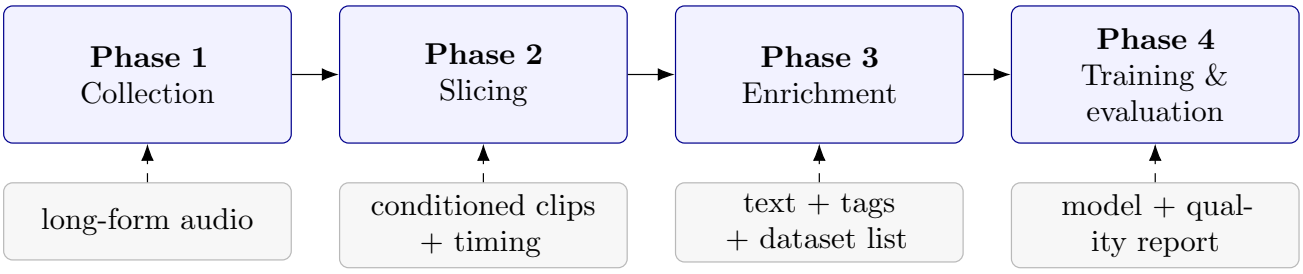


Figure 1: The four-stage corpus pipeline. Stage 2 is the subject of the companion Phase 2 document; this introduction describes the problem and the interfaces shared by all stages.

3.1 Collection

Collection acquires the source material and its provenance: scripts, speaker identity, language, licensing, and any existing annotations. The output is a set of long-form recordings, not training utterances. This stage determines the ceiling on corpus quality; no later stage can recover a clipped recording or an unknown speaker.

3.2 Slicing

Slicing is the first stage after collection. It turns long-form audio into the intervals that become training units and materializes them with timing and quality telemetry. The method, the candidate strategies, and the benchmark are specified in the companion Phase 2 document; this introduction relies only on its input and output contract.

3.3 Enrichment

Enrichment attaches meaning to each clip. Automatic speech recognition produces the transcript and word-level timing, a rule-based or learned classifier assigns an emotion or prosody tag, characters-per-second is computed as a quality signal, and the corpus is packaged in a line-oriented dataset list that a trainer can consume directly. Enrichment may reject a slice, for example because ASR produced no words, but it never changes the start and end offsets established upstream.

3.4 Training and evaluation

The final stage trains a TTS model and evaluates synthesized output with perceptual and speaker-identity metrics such as mean opinion score (MOS), speaker-embedding cosine similarity (SECS), and character error rate (CER). These metrics answer a different question from the corpus-level metrics of earlier stages, and they belong to the model rather than to the corpus. Keeping them in a separate phase prevents an unstable synthesis score from being mistaken for a segmentation defect.

3.5 The data contract

Across all four stages, an utterance is a tuple

$$u = \big(\text{audio},\ \text{text},\ \text{speaker},\ \text{language},\ \text{tags},\ \text{timing},\ \text{quality}\big),$$

where *timing* contains the original start and end offsets and *quality* contains peak level, RMS level, clipping rate, edge metrics, and a boundary status. The exported dataset list follows the widely used `path|speaker|language|text` convention. Every stage must either preserve this tuple or enrich it; no stage may silently change the meaning of a timestamp. This contract is what allows the phases to be developed, benchmarked, and documented independently.

The running example in this document and in the Phase 2 companion is 3.42 h (12,301 s) of Japanese in-game dialogue from a visual-novel source, publicly available at https://youtu.be/31Ju6z28IIs. The corpus is deliberately difficult: it contains music, emotional delivery, short interjections, and long pauses, which makes the failure modes above observable rather than hypothetical.

4 Design principles across stages

4.1 One contract, many transformations

The pipeline is a sequence of transformations over the same time base. The first stage selects which intervals exist; enrichment changes what is known about them; training changes nothing about them. Coordinate integrity is therefore a global invariant, not a stage-specific convenience. When a model internally resamples to 16 kHz, the mapping back to the original rate must be explicit, exact up to rounding, and clamped to the valid range.

4.2 Structural invariants

Six invariants are enforced by the corpus pipeline and respected downstream:

I1 Coordinate integrity. Every public boundary is an index in the original sampling rate.

I2 Non-overlap by construction. Consecutive intervals satisfy $b_k \le a_{k+1}$; contested padding is split at the midpoint.

I3 Duration contract. No interval exceeds the ceiling; intervals below the hard floor are dropped or handled by an explicit short-clip policy.

I4 Conditioning is terminal. Fades and normalization never influence boundary search.

I5 Isolation. Each strategy writes to its own namespace, and a failure in one does not abort the others.

I6 Reproducibility. DSP paths are deterministic; neural paths are deterministic for a fixed model, backend, device, and driver.

4.3 Evaluation as a first-class concern

A single number cannot rank a corpus pipeline, because the objectives are partly orthogonal. Boundary cleanliness, segmentation integrity, agreement with reference boundaries, and computational cost answer four different questions. The first-stage benchmark measures all four independently and reports their joint distribution. Later phases add their own axes—transcript accuracy, tag agreement, MOS, speaker similarity—without redefining the earlier ones.

4.4 Modularity and document boundaries

Each phase is documented separately and exposes a small interface to its neighbours. The first stage maps a waveform and a sampling rate to a boundary list; the enrichment stage consumes clips and offsets; the training stage consumes the packaged dataset. This separation is also a documentation decision: a reader who wants the segmentation method does not have to read the training methodology, and a future phase can be added without rewriting the introduction.

5 Scope, document map, and roadmap

This introduction covers the problem, the four-stage pipeline, the design principles shared by the stages, and the related literature. It intentionally does not derive the physics of sound, the mathematics of the Fourier transform, or the internals of TTS architectures. Neural components are treated as pre-trained inference engines; training those components is out of scope.

The phase documents are organized as follows:

- **Phase 2 — Slicing**: interval proposal, signal conditioning, evaluation, and the full benchmark. This is the companion document to the present introduction.
- **Phase 3 — Enrichment**: ASR transcription, emotion and prosody tagging, quality control, and dataset packaging. It will reuse the data contract defined here.
- **Phase 4 — Training and evaluation**: model training and perceptual or speaker-identity evaluation of synthesized speech.

The roadmap is deliberate: each phase has a stable input and output, so a change in one phase can be measured against the others without restructuring the project.

6 Conclusion

Corpus construction for TTS is not a single segmentation problem. It is a pipeline of transformations over a shared data contract, and each transformation has its own failure modes, constraints, and metrics. This document has introduced the four stages, the four corpus-level failure modes, the invariants that keep timestamps meaningful, and the literature on VAD, ASR timing, TTS corpora, and segmentation evaluation. The companion Phase 2 document turns the first stage into a concrete method and a reproducible benchmark; later documents will do the same for enrichment and model evaluation.

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